NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

HUMIDITY CABINET; OPERATION OF

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification is drawn to present detailed requirements for the apparatus, materials, and procedure to be used in the humidity cabinet corrosion test of materials intended as temporary type corrosion preventives. The humidity cabinet described herein is intended to produce a moisture saturated atmosphere with continuous condensation on the test specimens.

2. APPLICABLE SPECIFICATION

2.1 **Specification.**—The following specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

AIR FORCE-NAVY AERONAUTICAL SPECIFICATION

AN–C–121—Cloth; Mercerized Cotton, Airplane.

(*Army.*—Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of National Military Establishment specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va.) Copies of Air Force-Navy Aeronautical (AN) specifications may be obtained upon application to the Commanding Officer, U. S. Naval Air Development Station, Johnsville, Pa. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Air Force.*—Copies of National Military Establishment and Air Force-Navy Aeronautical (AN) specifications may be obtained upon application to the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 **General.**—The equipment for the humidity cabinet test shall be constructed and arranged to provide continuous, controlled operation during the test.

3.2 **Test room.**—The humidity cabinet test room shall be maintained at a temperature of 24° ±60 C. (75° ±10° F.) The humidity cabinet shall be situated in such a manner that exposure to air contaminated with foreign components such as hydrogen sulfide, sulfur dioxide, acids, etc., will be avoided.

3.3 **Construction.**—The cabinet shall conform to figures 2, 3, and 4.

3.4 **Temperature control.**—Adjustment shall be provided to control the air temperature throughout the panel exposure area at 49° ±1° C. (120° ±2° F.)

3.5 **Heating elements.**—Two 500-watt immersion heating elements shall be provided to heat the distilled water in the cabinet. Thermostats and pilot lights shall be provided in the heating circuit as shown in wiring diagram of figure 4. The intermittent thermostat shall be set to maintain the temperature at 49° ±1° C. (120° ±2° F.) at the test panels in the cabinet. The auxiliary thermostat shall be set to break contact a few degrees above the setting of the intermittent thermostat. The pilot light across the auxiliary serves as a warning light if temperature of cabinet exceeds the thermostat settings. The switch on the auxiliary heating element is normally open after cabinet has been brought up to proper temperature and during operation of the cabinet. The heating elements and thermostatic devices shall be located as shown in figure 2.

3.6 **Cabinet cover.**—The top of the exposure chamber shall consist of a corrosion resistant metal frame which shall be hinged along one side so that top, when closed, provides a seal along the top edges of the exposure chamber. Two thicknesses of desized cotton cloth conforming to Specification AN–C–121 (see 6.2) shall be tautly secured to the frame as indicated on figure 2. The cloth is intended to prevent drippage of condensed moisture onto the test specimens.

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3.7 Air supply.—The air supply shall be capable of delivering sufficient air to maintain a constant flow of air through the cabinet as specified herein and free from oil and industrial fumes as sulfur dioxide. Oil or foreign particles present in the air after the air compressor, shall be removed by passing the air through a tower lightly packed with asbestos or glass wool. The air line connection from the tower to the cabinet shall be clean and as short as possible in order to avoid small particles of dirt, scale, or other foreign matter.

3.8 Distilled water.—Distilled water shall be used in the test cabinet and level maintained as indicated on figure 2 or shall consist of a suitable float actuated water supply device. If a constant source of distilled water is not available, a glass water storage bottle placed above water level in cabinet, with glass syphon tube and hydrostatic leveling tube, may be used.

3.9 Instruments.—

3.9.1 *Temperature indicating and/or recording devices.*—Suitable partial immersion thermometers, temperature indicating or continuous recording devices with subdivisions of 0.5° C. (0.9° F.) and a scale error of not more than 0.5° C. (0.9° F.) shall be provided for denoting temperatures during cabinet operation.

3.9.2 *Air metering device.*—A suitable air metering device to indicate or continuously record the air flow shall be provided.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection.—The equipment, during and after manufacture, and the preparation for delivery thereof shall be subject to inspection by Government inspectors who shall be afforded proper facilities for determining conformance with requirements of this specification.

4.2 Preparation of test specimens.—Size of specimen exposure panels and location of holes shall be as shown on figure 1. Specimens for test shall be prepared as required in detail specifications applicable to the individual material under test.

4.3 Test procedures.—

4.3.1 *General.*—The cabinet shall be maintained in continuous operation at a temperature of 49°±1° C. (120°±2° F.) in the area occupied by the test specimens. The insertion or removal of test specimens shall be so regulated as to limit the interruption of test conditions to a minimum.

4.3.2 *Arrangement of specimens.*—Panels shall be suspended as shown on figure 2 by means of stainless steel or monel hooks. The distance between test specimens shall be not less than the distance between notches on rotating stage shown on ngure 3. Blank stainless steel panels shall be suspended from all notches unoccupied by test specimens and from rotating stage support arms so as to maintain equal distances between panels.

4.3.3 *Air flow.*—The conditioned air shall pass through the cabinet at 8±1 linear feet per hour. (8±1 linear feet per hour for cabinet shown on fig. 2 is equivalent to 31.2 plus 4 or minus 3.8 cubic feet per hour of bone dry air at 70° F. and 1½ p. s. i.) Throughout the test period condensation shall take place on the test panels. This condition is of prime importance and every cabinet should be periodically checked for this condensation on the test specimens.

4.3.4 *Operating records.*—Rate of air flow, and cabinet temperatures shall be observed and recorded at least once during each 24-hour period.

4.4 Evaluation of results.—At the end of the test period the specimens shall be carefully removed, treated, and evaluated in accordance with the applicable specification for the material under test.

4.5 Maintenance.—

4.5.1 *Care of air supply.*—The tower packed with asbestos or glass wool for removing oil and foreign matter from the air supply shall be replaced or cleaned whenever required.

4.5.2 *Maintenance of humidity cabinet.*—The cabinet shall be cleaned and distilled water replaced at least once every 60 days to remove any contamination which may be introduced from the test panels themselves.

5. PREPARATION FOR DELIVERY

5.1 This specification does not include requirements coming within the scope of this particular section.

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6. NOTES

6.1 Invitation for bids and contracts or orders should specify the title, number and date of this specification.

6.2 Desizing cotton cloth.—A suitable method for desizing cloth is described by A. S. T. M. Method D629–42T.

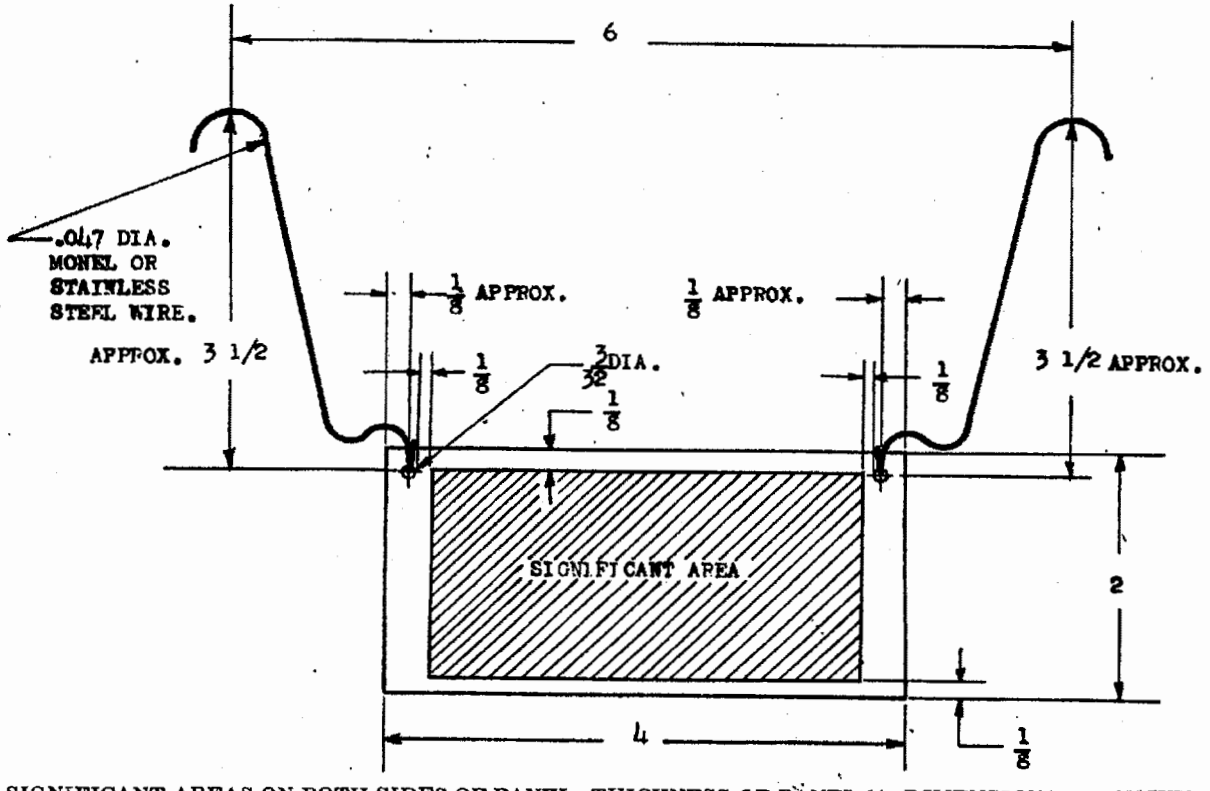
6.3 The Precision Scientific Co., 1750 North Springfield Avenue, Chicago, Ill., has produced a humidity cabinet meeting the requirements of this specification.

6.4 When used to evaluate corrosion preventives, this cabinet is considered a satisfactory means to differentiate between ineffective and reasonably good protective compounds for uses in which conditions of high humidity and condensation are encountered. This may not apply, however, for conditions in which no condensation, actual rainfall, or water immersion are encountered.

6.5 It has not been ascertained whether the difficulties encountered in obtaining reproducible results are attributable to the design and operation of the cabinet, or to the preparation of test specimens and evaluation of test results as have been described in applicable documents, or to a combination of the two.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

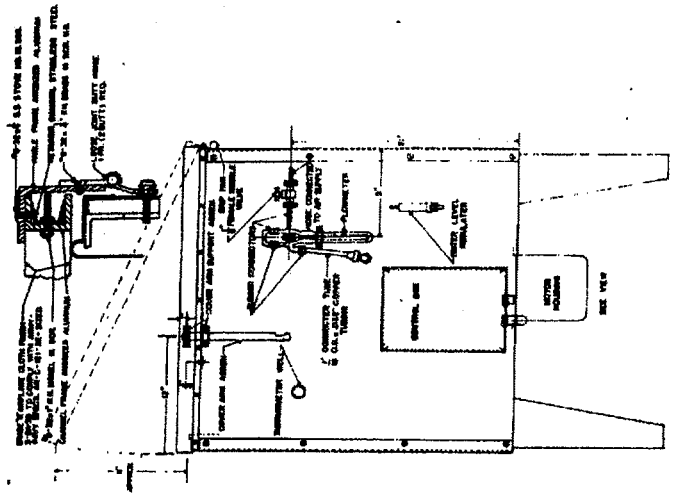
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Air Force.
Other interest:
Army—COQST.
Navy—AMOShS.

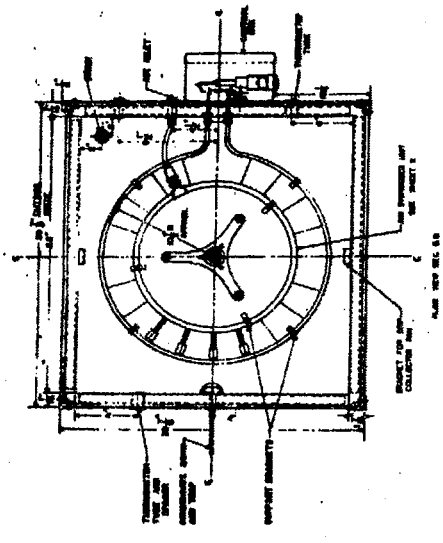


SIGNIFICANT AREAS ON BOTH SIDES OF PANEL. THICKNESS OF PANEL ⅛. DIMENSIONS IN INCHES.

FIGURE 1.—*Humidity cabinet exposure panel and suspension hooks.*

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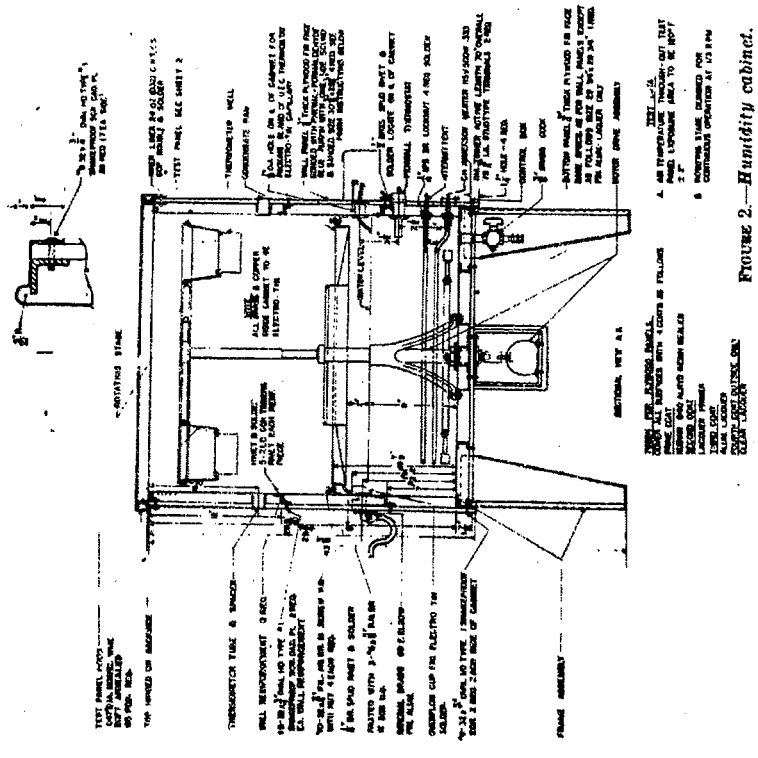


FIGURE 2.—*Humidity cabinet.*

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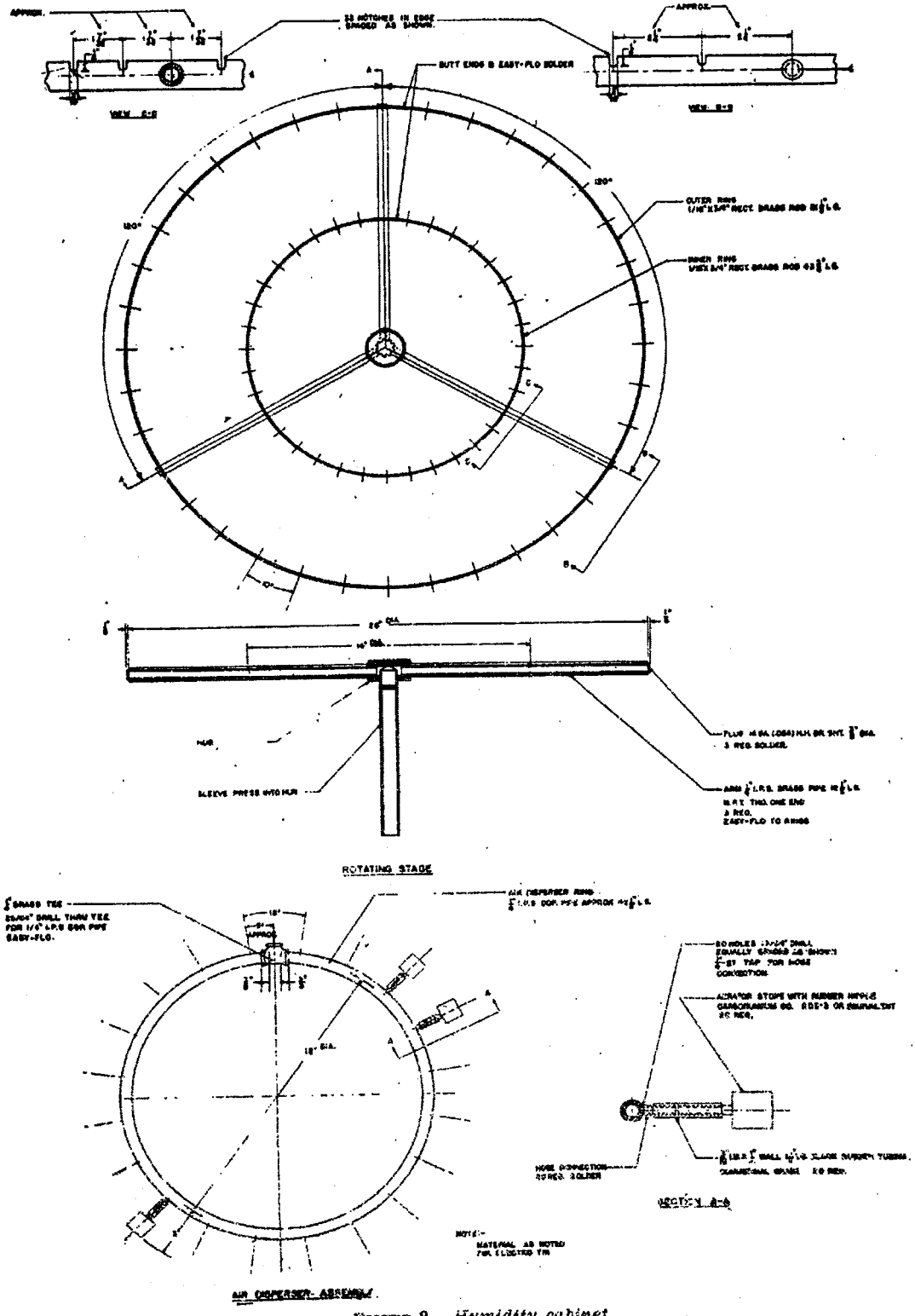


FIGURE 8.—*Humidity cabinet.*

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